

## ASCII MASTER FLOW – PANEL 1/12: Aircraft-to-system terminology ladder

UAV -> aircraft only

UAS -> aircraft + remote station + C2 + associated elements

RPAS -> remotely piloted UAS; pilot in command

DRONE -> colloquial umbrella

RULE -> unmanned does not mean autonomous

## ASCII MASTER FLOW – PANEL 2/12: Integrated UAS architecture

AIRFRAME + POWER + PROPULSION

FLIGHT CONTROLLER + NAVIGATION SENSORS

COMMAND / TELEMETRY / PAYLOAD-DATA LINKS

REMOTE PILOT STATION OR ONBOARD LOGIC

PAYLOAD -> mission capability

## ASCII MASTER FLOW – PANEL 3/12: Flight-control feedback loop

COMMAND / PATH -> desired state  
IMU / SENSORS -> measured state  
CONTROLLER -> compare + correct  
ROTORS / CONTROL SURFACES -> physical response  
NEW STATE -> sensed again

## ASCII MASTER FLOW – PANEL 4/12: Platform category comparison

MULTIROTOR -> hover + confined operation | rotor-borne flight

FIXED-WING -> wider-area coverage | launch/recovery need

HYBRID VTOL -> vertical access + wing cruise | integration complexity

PAYLOAD CHANGES -> mass + power + data + control burden

NO LABEL -> universal range/endurance guarantee

## ASCI MASTER FLOW – PANEL 5/12: Human-control continuum

TELEOPERATION -> human directly commands

AUTOMATION -> fixed sequence

AUTONOMY -> senses and selects actions

AI SUBSYSTEM -> perception/decision support, not legal authority

ACCOUNTABILITY -> human oversight + defined fallback

## ASCII MASTER FLOW – PANEL 6/12: Swarm coordination mesh

MEMBERS -> multiple aircraft

LOCALISATION -> self and neighbour state

COMMUNICATION -> shared task/state information

CONTROL RULES -> allocation + formation + adaptation

FAILURE TEST -> link loss or member loss

## ASCII MASTER FLOW – PANEL 7/12: VLOS to BVLOS assurance bridge

VLOS -> pilot maintains unaided visual contact

BVLOS -> aircraft beyond visual envelope

BRIDGE -> reliable C2 + navigation + detect-and-avoid

CONTINGENCY -> lost link + diversion + termination logic

STATUS -> trial authorisation != routine permission

## ASCII MASTER FLOW – PANEL 8/12: Counter-UAS decision funnel

DETECT -> radar | RF | acoustic | optical

IDENTIFY / TRACK -> classify and maintain picture

DECIDE -> authority + proportionality + surroundings

MITIGATE -> electronic | capture | other authorised response

LIMIT -> jamming/spoofing can affect legitimate systems



## ASCII MASTER FLOW – PANEL 9/12: Civilian application evidence grid

AGRICULTURE -> monitor / map / spray

SCIENCE -> volcano / wildlife observation

GOVERNANCE -> survey / inspection / disaster assessment

LOGISTICS -> pilot or authorised mission, not blanket service

EVIDENCE -> imagery + interpretation + field verification

## ASCII MASTER FLOW – PANEL 10/12: Safety privacy and liability shield

ENGINEERING -> airworthiness + maintenance + failsafes

OPERATIONS -> pilot + weather + route + payload discipline

DATA -> purpose + access + retention + geospatial sensitivity

CYBER -> C2/navigation/payload-link resilience

LIABILITY -> trace operator/software/manufacturer function

MUST REMEMBER: Drones combine airframe, propulsion, control, communication/navigation,...

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## ASCII MASTER FLOW – PANEL 11/12: Indian governance responsibility map

MOCA -> policy / notification

DGCA -> civil regulation / certification

AAI / ATC -> airspace and traffic services

BCAS -> aviation security

DIGITAL SKY -> workflow platform, not ATC

CLOSE DISTINCTION: Drone is not necessarily autonomous, remote piloting is not AI,...

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## ASCII MASTER FLOW – PANEL 12/12: Approval-to-outcome status rail

RULE / AMENDMENT -> legal text

TYPE CERTIFICATE / UIN / RPC -> distinct compliance conditions

TRIAL / SCHEME / INCENTIVE -> authorised or supported route

PROCUREMENT / DELIVERY -> not operational deployment

DEPLOYMENT -> not verified safety, scale or outcome

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